Ibex GP Progress

(Zero-RISC)

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Outlines

- NDM updates
- Floor planning
- Power planning
- Inquiries

NDM Updates

Synthesis Rerun

- Rerun synthesis using the same .db used in the PDK NDM.

```
----------------------------------------------------------------------------------
                              LIBRARY INFORMATION
----------------------------------------------------------------------------------

Search path : .

Units :
    time                   : 1.00ns
    resistance             : 1.00MOhm
    capacitance            : 1.00fF
    voltage                : 1.00V
    current                : 1.00uA
    power                  : 1.00pW

Tech file : /mnt/hgfs/0_GP/PDK/saed14_pdk/SAED14nm_PDK_06052019/SAED14nm_PDK_12232018/techfiles/saed14nm_1p9m_mw.tf

Number of active scenarios      = 1
Number of inactive scenarios    = 0

Total number of standard cells  = 1012

Total number of dont_use lib cells      = 0
Total number of dont_touch lib cells    = 0

Total number of buffers         = 91
Total number of inverters       = 51
Total number of flip-flops      = 172
Total number of latches         = 45
Total number of ICGs            = 0


Library : saed14hvt_frame_timing_ccs
  File path : /mnt/hgfs/0_GP/PDK/saed14_pdk/SAED14nm_EDK_CORE_HVT_v_062020/stdcell_hvt/ndm/saed14hvt_frame_timing_ccs.ndm
  Source .db libs :
    saed14hvt_tt0p8v25c : /SCRATCH/armanga/ndm_gen/liberty/logic_synth/saed14rvt_tt0p8v25c.db
1
icc2_shell>
```

Synthesis QoR

```
Timing Path Group 'cluster_clock_gating'
-----------------------------------
Levels of Logic:                  71.00
Critical Path Length:              2.58
Critical Path Slack:               2.77
Critical Path Clk Period:          6.25
Total Negative Slack:              0.00
No. of Violating Paths:            0.00
Worst Hold Violation:            -61.93
Total Hold Violation:          -4319.06
No. of Hold Violations:         2015.00
-----------------------------------
```

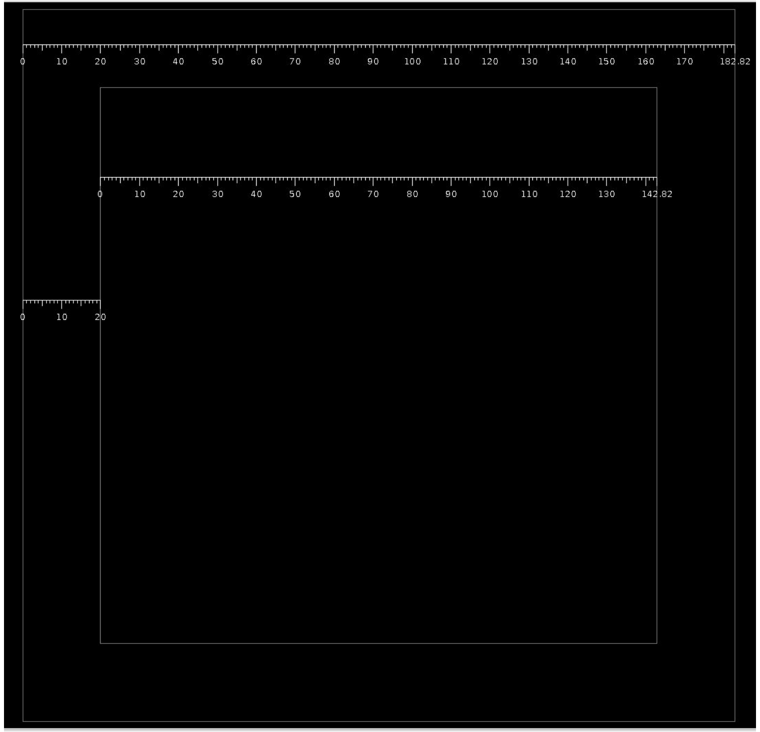
```
Area
-----------------------------------
Combinational Area:        5302.469977
Noncombinational Area:     2193.848432
Buf/Inv Area:               562.459183
Total Buffer Area:              264.80
Total Inverter Area:            332.16
Macro/Black Box Area:         0.000000
Net Area:                 10470.425709
-----------------------------------
Cell Area:                 7496.318409
Design Area:              17966.744118
```

Floor Planning

Initialize Floorplan

```
initialize_floorplan -core_utilization 0.6 -core_offset 20 -shape R -flip_first_row true -control_type die
```

- Die side length = 182.02
- Core side length = 142.02
- Core offset = 20



Pin Placement

Group pins by their function. For example, SIDE_2 is a list that contains all the ports communicating with the memory such as data_wdata_o & data_rdata_i

```
set_block_pin_constraints -self -allowed_layers M6 -pin_spacing 1 -sides {1} -corner_keepout_num_tracks 1
place_pins -ports [get_ports $SIDE_1]
place_pins -ports [get_ports $OUT_PORT_1BIT]

set_block_pin_constraints -self -allowed_layers M7 -pin_spacing 1 -sides {2} -corner_keepout_num_tracks 1
place_pins -ports [get_ports $SIDE_2]
place_pins -ports [get_ports $IN_PORT_1BIT]

set_block_pin_constraints -self -allowed_layers M6 -pin_spacing 1 -sides {3} -corner_keepout_num_tracks 1
place_pins -ports [get_ports $SIDE_3]

set_block_pin_constraints -self -allowed_layers M7 -pin_spacing 1 -sides {4} -corner_keepout_num_tracks 1
place_pins -ports [get_ports $SIDE_4]
```

Pin Placement

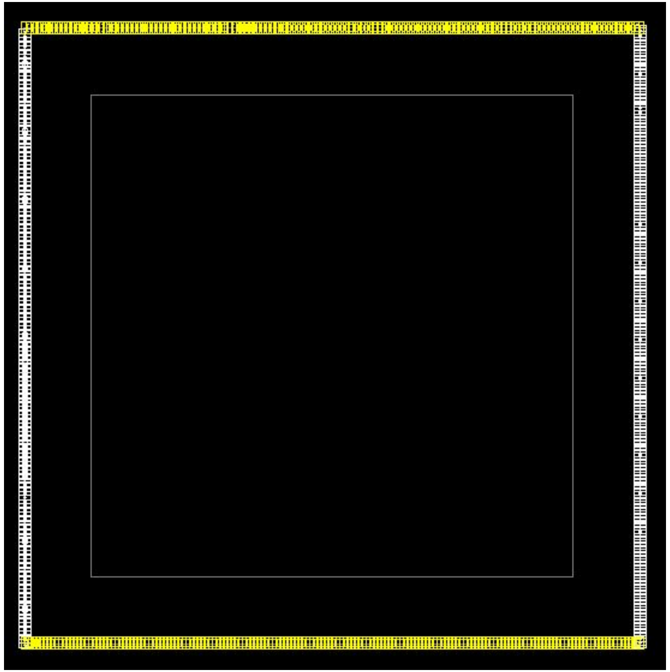
Not equally distributed on all sides as we grouped pins by their function.

Side 1 has 154 ports.

Side 2 has 147 ports.

Side 3 has 192 ports.

Side 4 has 164 ports.



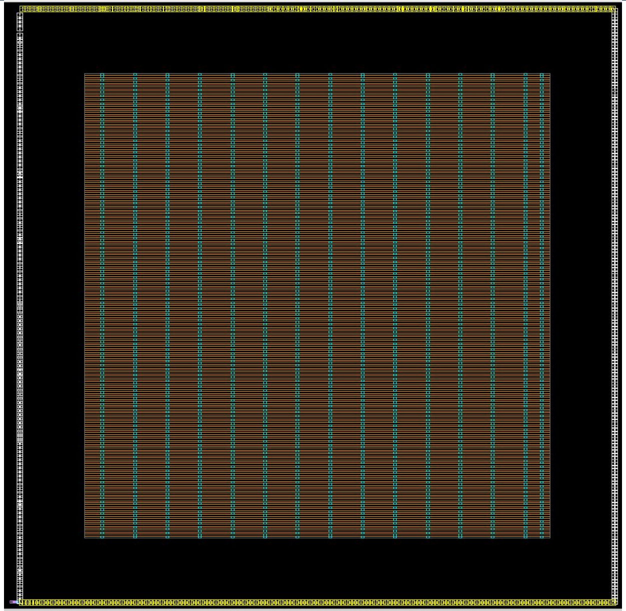
Pin Placement

Group pins by their function. For example, SIDE_2 is a list that contains all the ports communicating with the memory such as data_wdata_o & data_rdata_i

```
set_block_pin_constraints -self -allowed_layers M6 -pin_spacing 1 -sides {1} -corner_keepout_num_tracks 1
place_pins -ports [get_ports $SIDE_1]
place_pins -ports [get_ports $OUT_PORT_1BIT]

set_block_pin_constraints -self -allowed_layers M7 -pin_spacing 1 -sides {2} -corner_keepout_num_tracks 1
place_pins -ports [get_ports $SIDE_2]
place_pins -ports [get_ports $IN_PORT_1BIT]

set_block_pin_constraints -self -allowed_layers M6 -pin_spacing 1 -sides {3} -corner_keepout_num_tracks 1
place_pins -ports [get_ports $SIDE_3]

set_block_pin_constraints -self -allowed_layers M7 -pin_spacing 1 -sides {4} -corner_keepout_num_tracks 1
place_pins -ports [get_ports $SIDE_4]
```

Tap Cells

```
# Tap cells
create_tap_cells -lib_cell [get_lib_cell */SAEDRVT14_TAPDS] -pattern stagger -distance 20
```



Is that enough?

How to remove tap cells? If undo is required.

Power Planning

Create Ports

```
## -- Create VDD & VSS ports - if designing a chip not IP (to be confirmed)
create_port -port_type ground -direction in GND
create_port -port_type power -direction in VDD
```

Because we are planning a chip but what about the IP case?

Create Nets

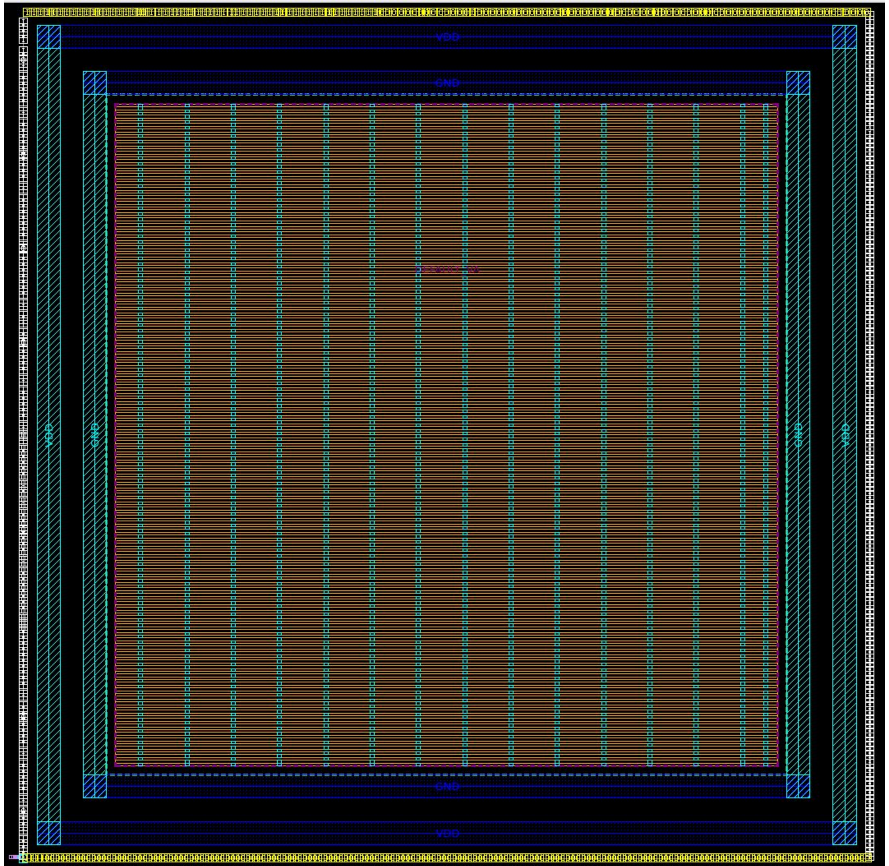
```
## -- Create power & ground nets
create_net -ground GND
create_net -power VDD


connect_net -net GND [get_ports GND]
connect_net -net VDD [get_ports VDD]


connect_pg_net -net VDD [get_pins -hierarchical */VDD]
connect_pg_net -net GND [get_pins -hierarchical */VSS]
```

PG Ring

```
# Create region for the PG ring
# 4 sides because it's an R shape. Pay attention if another shape is used such as U.
create_pg_region $PG_REGION_NAME -core \
                -expand_by_edge "{{side: 1} {offset: $PG_TOP_OFFSET}} {{side: 2} {offset: $PG_TOP_OFFSET}} \
                                 {{side: 3} {offset: $PG_TOP_OFFSET}} {{side: 4} {offset: $PG_TOP_OFFSET}}"

# Create PG ring pattern
create_pg_ring_pattern \
                 ring \
                 -horizontal_layer $PG_H_LAYER -vertical_layer $PG_V_LAYER \
                 -horizontal_width $PG_H_WIDTH -vertical_width $PG_V_WIDTH \
                 -horizontal_spacing $PG_H_SPACING -vertical_spacing $PG_V_SPACING


# Set strategy
set_pg_strategy  ring -pg_regions "$PG_REGION_NAME" -pattern {{ name: ring} { nets: "GND VDD" }}

compile_pg -strategies ring
```

```
set PG_TOP_OFFSET 2
set PG_REGION_NAME "Power_Ring"
set PG_H_LAYER M8
set PG_V_LAYER M9
set PG_H_WIDTH 5
set PG_V_WIDTH 5
set PG_H_SPACING 5
set PG_V_SPACING 5
```

PG Ring

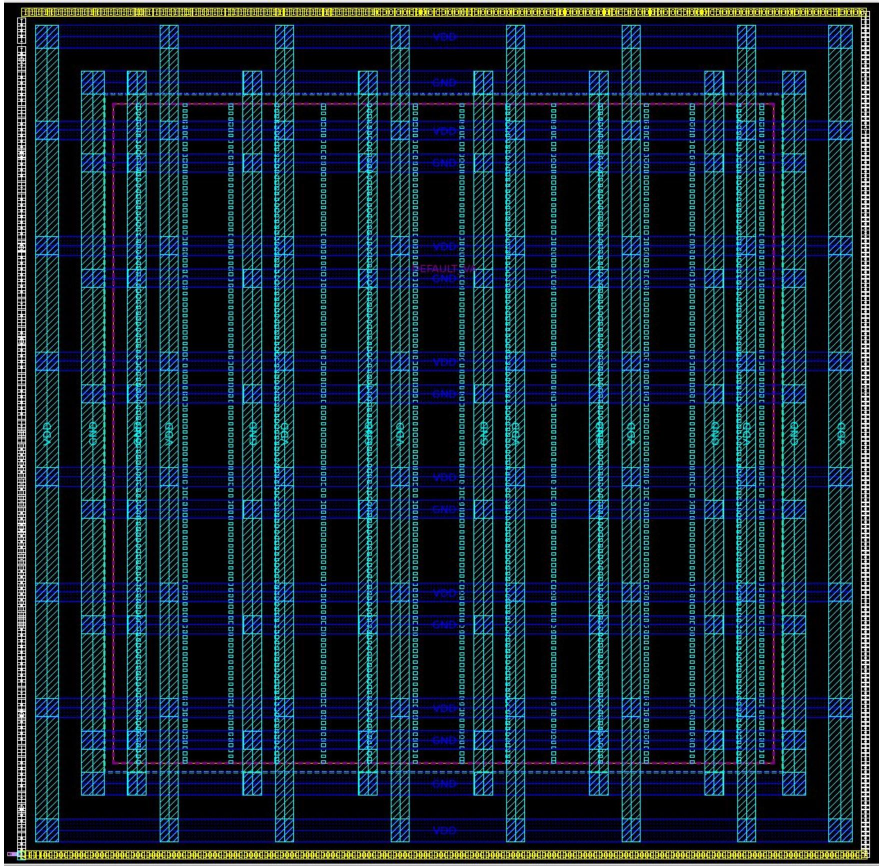


PG Mesh

```
create_pg_mesh_pattern \
                mesh \
                -parameter {MESH_H_LAYER MESH_H_WIDTH MESH_H_SPACING MESH_H_PITCH MESH_H_OFFSET\
                            MESH_V_LAYER MESH_V_WIDTH MESH_V_SPACING MESH_V_PITCH MESH_V_OFFSET} \
                -layers { \
                        { {horizontal_layer: @MESH_H_LAYER} {width: @MESH_H_WIDTH} {spacing: @MESH_H_SPACING} {pitch: @MESH_H_PITCH} {offset: @MESH_H_OFFSET} {trim : true} } \
                        { {vertical_layer: @MESH_V_LAYER}   {width: @MESH_V_WIDTH} {spacing: @MESH_V_SPACING} {pitch: @MESH_V_PITCH} {offset: @MESH_V_OFFSET}  {trim : true} } \
                        }

set_pg_strategy mesh \
                -core \
                -pattern {{name: mesh} {nets: {"GND VDD"}} \
                                     {parameters: "$MESH_H_LAYER $MESH_H_WIDTH $MESH_H_SPACING $MESH_H_PITCH $MESH_H_OFFSET \
                                                   $MESH_V_LAYER $MESH_V_WIDTH $MESH_V_SPACING $MESH_V_PITCH $MESH_V_OFFSET"}} \
                -extension { {stop:outermost_ring} }


compile_pg -strategies mesh
```

```
set MESH_H_LAYER "M8"
set MESH_V_LAYER "M9"
set MESH_H_WIDTH 4
set MESH_V_WIDTH 4
set MESH_H_SPACING 3
set MESH_V_SPACING 3
set MESH_H_PITCH 25
set MESH_V_PITCH 25
set MESH_H_OFFSET 5
set MESH_V_OFFSET 5
```

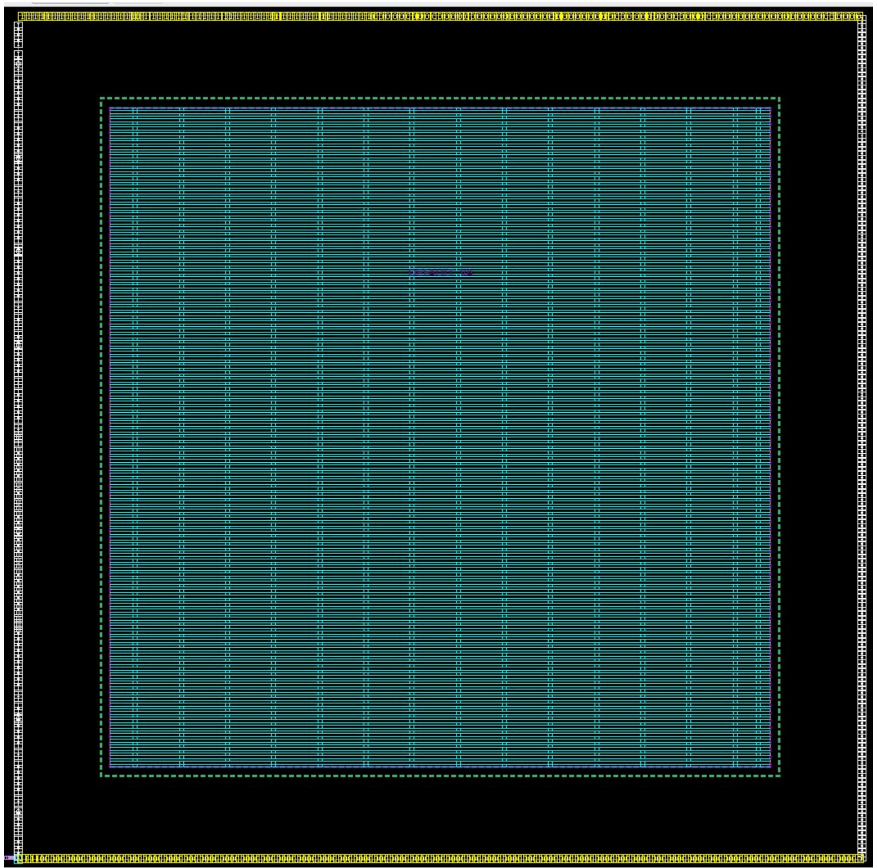
PG Mesh



Cell Rails

```
# 0.094 is the width if M1 used in SAED14 std cells
set CELL_RAIL_WIDTH 0.094
set CELL_RAIL_LAYER M1

create_pg_std_cell_conn_pattern cell_rail -rail_width $CELL_RAIL_WIDTH -layers $CELL_RAIL_LAYER

set_pg_strategy cell_rail -pattern {{name: cell_rail} {nets: "GND VDD"}} -core

compile_pg -strategies cell_rail
```

Cell Rails

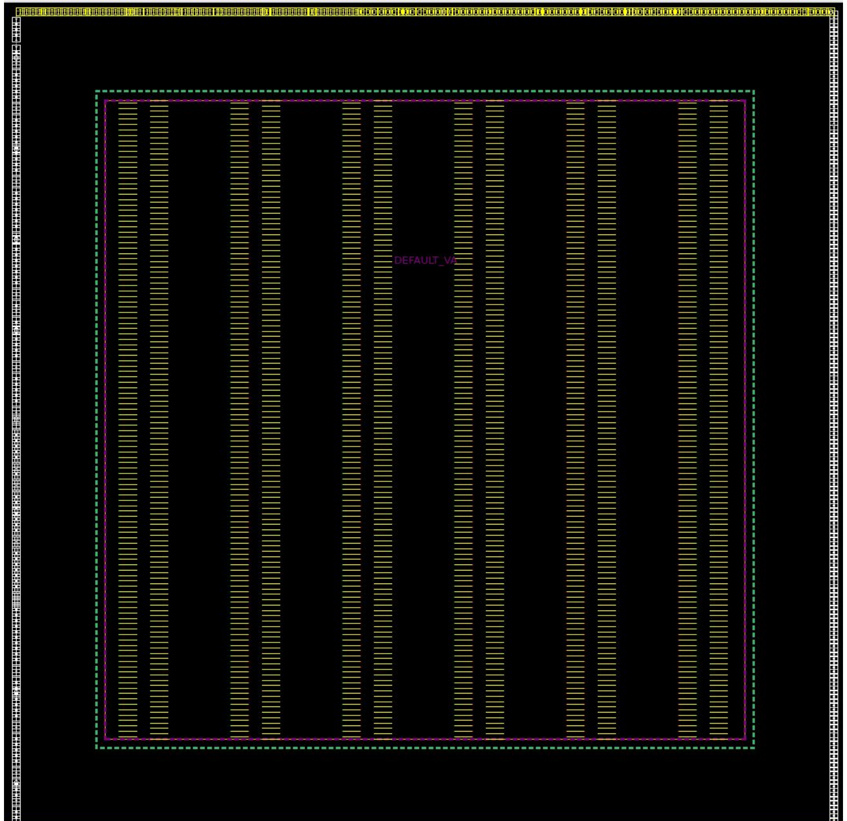


Force Vias

```
# Vias aren't placed automatically between the mesh layers and the lower layers to not violate PG DRC but they are needed indeed.
create_pg_vias -nets "GND VDD" -from_layers $MESH_V_LAYER -to_layers $CELL_RAIL_LAYER -drc no_check -pitch {5 5}

```

Vias are not created automatically between the mesh layers and the cell rails because they violate PG DRC so, we create them while disabling DRC check.

Force Vias



PG DRC

ErrorSet	Total
ibex_1_rvt:ibex_1_rvt_powerplan.design	30432
DRC_report_by_check_pg_drc	30432
Cut-To-Cut Spacing	1080
Error between diff-layers cuts	25812
Metal Spacing	48
Overlap of Via-Cuts	3240
Short	252

Inquiries

Q 1

What is this used for?

```
> set_wire_track_pattern -site_def unit -layer M1 -mode uniform \
        -mask_constraint {mask_two mask_one} -coord 0.037 \
        -space 0.074 -direction vertical
```

No manual found for such command!

From Eng. Karim Hussein’s script.

Q 2

When we should place the IO pads and how?

Should we specify a margin for them inside the core offset region? If so, what about the PG ring?

Thank you.